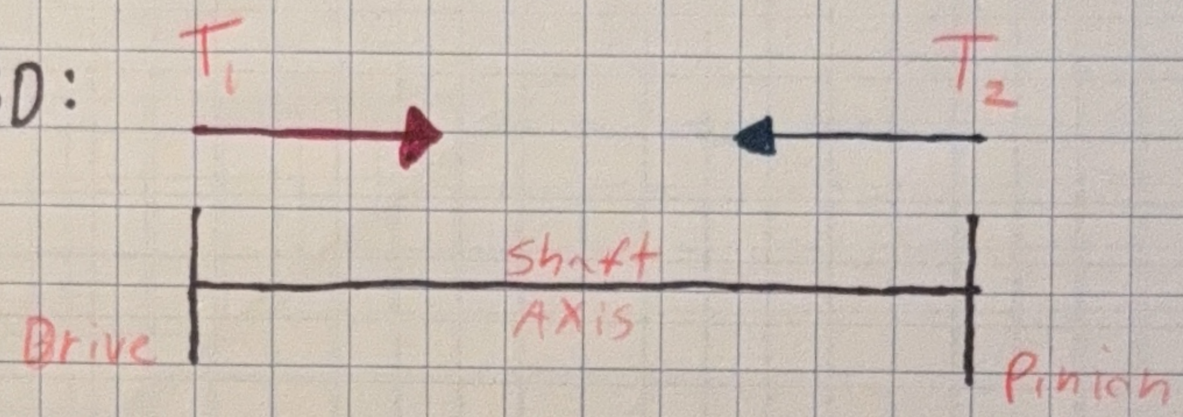
Input Calculations

Speed: $N_{in} = N_{out}\frac{48}{18} =$ [redacted] **2933.33 RPM**

Torque: $T_{in} = \frac{T_{out} \cdot N_{out}}{N_{in}} = (1.25)(1100)/(2933.33) =$ **0.46875 in·lbf**

FBD:



$T_1 = T_2 \quad ; \Sigma T = 0$

$$\tau_{max} = \frac{Tr}{J}, \quad r = \frac{d}{2}, \quad J = \frac{\pi d^4}{32}$$

$$\tau_{max} = \frac{Td}{2}\left(\frac{32}{\pi d^4}\right) = \frac{16T}{\pi d^3}$$

$$\tau_{max} = \frac{16(0.46875)}{\pi(0.125)^3}$$

$\tau_{max} = 1222.31$ psi

$1222.31 \text{ psi} < 4000 \text{ psi}$